
THE E-WASTE LAW DISCONNECT: CHALLENGES IN TACKLING ELECTRONIC WASTE IN THE PHILIPPINES

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Connection, work productivity, effective learning, convenience, and, most prominently, entertainment—a short list of the diverse uses of our gadgets, seamlessly integrating into nearly every aspect of modern life. Gadgets have become indispensable tools from enabling instant communication across the globe to providing access to information and enhancing personal productivity. They support learning, offer fitness insights, simplify home management, and bring endless entertainment options to our fingertips. These benefits, however, also bring challenges like the environmental impact of electronic waste.

But how do we truly cut off our old gadgets? Have we ever considered the impact on how we dispose of them?

In my honest opinion, I do not give it much thought. I often toss them aside, thinking they are just pieces of junk with no further use, without regard to the fact that I have contributed to the growing e-waste crisis. I know that I am not the only one guilty of such acts. These small contributions are aggregated and form part of the global crisis that we are currently facing and addressing the conservation of our environment for the present and future generations.

According to the UN's fourth Global E-waste Monitor (GEM), electronic waste (e-waste) generation is surging, with a record of 62 million tons produced in 2022 which contributes to an increase of 82% since 2010. This figure is projected to rise to 82 million tons by 2030. Alarmingly, only 22.3% of e-waste was properly collected and recycled in 2022, leaving about \$62 billion worth of recoverable resources unaccounted for and heightening pollution risks globally.¹

The Philippines is facing a growing e-waste crisis, with e-waste reaching five hundred thirty-seven million (537,000,000) kilograms in 2022. This surge raises serious concerns about environmental and health risks. According to the latest UN Global E-waste Monitor, the country is among the top producers of e-waste in Southeast Asia.²

Before diving further, let us first define what e-waste is. Electronic waste, or e-waste, refers to all items of Electrical and Electronic Equipment (EEE) and its parts

¹ United Nations Institute for Training and Research; The Global E-waste Monitor 2024, <https://ewastemonitor.info/the-global-e-waste-monitor-2024/>

² Geneva Internet Platform, March 27, 2024, D Philippines faces escalating e-waste crisis | Digital Watch Observatory, <https://dig.watch/updates/philippines-faces-escalating-e-waste-crisis>

that have been discarded by its owner as waste without the intent of re-use. E-waste is also referred to as Waste Electrical and Electronic Equipment (WEEE), electronic waste, or e-scrap in different regions and under different circumstances in the world. It includes a wide range of products that consist of almost any household or business item with circuitry or electrical components with power or battery supply.³

Environmental Protection in the Philippines

The 1987 Philippine Constitution specifically embodied the protection of the environment in its provisions. Article 2, Sections 15 and 16 clearly state that the Constitution protects the right of the people to a balanced and healthful ecology.⁴ In his speech at the Judges Forum on Environmental Protection, former Justice Puno encouraged his fellow judges in disputes about environmental issues to equip themselves with a commanding armor of the emerging substantive body of Environmental Law, especially considering the depth and breadth of contemporary environmental issues. He also nudged them to be creative and to shed their fears even when they find themselves in some unfenced spaces of our environmental law.⁵

Former Justice Puno even acknowledged that the Philippine environmental laws are somehow inadequate. Times might have changed; however, it seems like we still lack the statute addressing electronic waste problems.

In 2000, the Philippine legislature enacted R.A. No. 9003 also known as the "Ecological Solid Waste Management Act of 2000," it aims to provide a systematic and comprehensive approach to waste management in the Philippines. The law establishes the framework for managing solid waste, emphasizing environmental protection and sustainability. It has established a national solid waste framework regarding the proper ways to collect and segregate solid waste, and provide public awareness and education. The law also gives incentives to encourage the participation of the general public and imposes penalties to its violators.⁶

While R.A. No. 9003 promotes general waste management practices such as reduction, reuse, and recycling, it falls short in actively engaging manufacturers and producers in the e-waste management process. A comprehensive approach is needed to hold these stakeholders accountable and ensure they contribute to mitigating the impacts of e-waste on the environment.

³ The Global E-waste Statistic Partnership, <https://globalewaste.org/what-is-e-waste/#:~:text=Electronic%20waste%2C%20or%20e-waste%2C%20refers%20to%20all%20items,owner%20as%20waste%20without%20the%20intent%20of%20re-use>

⁴ CONST. (1987), art. II, §15 and §16 (Phil.).

⁵ Philippine Environmental Law Practice and the Role of the Courts, Aug. 14, 2003, <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/2/4800>

⁶ Ecological Solid Waste Management Act of 2000, Rep. Act No. 9003 (Phil.).

E-waste should be addressed differently from other solid waste because it imposes a different level of threat both to the health and environment as evidenced by related studies. Generally solid waste is any unwanted or unusable material that is solid in its state of matter.⁷ Which is generally a broader category of waste, while electronic wastes are items that contain many different toxic substances⁸ that are discarded from electronic devices. Unfortunately, the Ecological Solid Waste Act of 2000 does not provide enough solutions to address the crisis being presented, a detailed and well-defined statute should be passed to address the issue.

Effects of E-waste

Various research shows the bad effects of e-waste, both on the environment and human body as it causes air pollution from burning e-waste, water pollution due to toxic chemicals, and soil contamination that affects agriculture. These pollutants can harm ecosystems and enter the food chain, posing risks to human health.⁹ E-waste is non-biodegradable and toxic, leading to accumulation in the environment, including soil, air, water, and living organisms. Practices like open-air burning and acid baths to extract valuable materials release harmful substances into the environment.¹⁰

One of the most common effects of E-waste on the environment is air pollution. E-waste can pollute the air when improperly shredded, melted, or burned. These wrong practices release dust particles or toxins, such as dioxins, into the environment and cause air pollution.¹¹ When the improper disposal of e-waste occurs in regular landfills or in places where it is dumped illegally, both heavy metals and flame retardants can leach directly from the e-waste into the soil, causing contamination of underlying groundwater or contamination of crops that may be planted nearby or in the area in the future.¹² After soil contamination, toxins can eventually make their way into nearby water. Heavy metals from improperly disposed e-waste, such as mercury, lithium, lead, and barium, can leach through the earth, even reaching the groundwater. Consequently, when these heavy metals reach groundwater, they eventually make their way into ponds, streams, rivers, and lakes.¹³ Its improper disposal affects both land and sea animals. It is also worth considering the effects that electronic waste has on climate change. Every device ever produced has a carbon footprint and is contributing to human-made global warming. The production of a ton

⁷ The Daily Eco, Jan. 8, 2003- What is Solid Waste, <https://www.thedailyeco.com/what-is-solid-waste-definition-and-examples-41.html>

⁸ Electronic Waste, Oct. 1, 2004, <https://www.who.int/news-room/fact-sheets/detail/electronic-waste-%28e-waste%29>

⁹ Wasted: The Effects of E-Waste on the Environment, Cleanlites Recycling, Jun. 22, 2019, <https://cleanlites.com/wasted-effects-of-ewaste>

¹⁰ GENEVA Network, The Growing Environmental Risks of E-Waste, (2024) <https://www.genevaenvironmentnetwork.org/resources/updates/the-growing-environmental-risks-of-e-waste/>

¹¹ Environment Go; 10 Environmental Impacts of Electronic Waste, Jan. 9, 2023, <https://environmentgo.com/environmental-impacts-of-electronic-waste/>

¹² *Id.*

¹³ *Id.*

of laptops emits up to ten metric tons of carbon dioxide. When the carbon dioxide released over a device's lifetime is considered, it predominantly occurs during production and usually before consumers buy a product.¹⁴ Agricultural land use is significantly impacted by soil and water pollution. Heavy metals and flame-retardant chemicals can stick around in crops. Livestock, wildlife, and human populations are at increased risk of consuming crops contaminated with these pollutants.¹⁵

E-waste presents serious environmental challenges, from polluting air, soil, and water to contributing to climate change. Improper disposal releases harmful chemicals and heavy metals, contaminating ecosystems and entering the food chain, which can have adverse effects on human health, agriculture, wildlife, generally, the environment.

The Proposed Bill

On July 22, 2018, Senator Francis Tolentino introduced Senate Bill No. 751¹⁶ which is the "E-Waste Management Act," and it aims to regulate e-waste disposal in the Philippines. It emphasizes the need for manufacturers to adopt recovery plans for electronic equipment and mandates convenient, accessible recycling options. Manufacturers must label products for tracking, set up collection points, and report yearly on waste management. The bill includes penalties for non-compliance, outlines roles for local governments, and allocates fines imposed to an "E-Waste Recycling Fund" to support its initiatives. It seeks to align with the existing ecological laws for a comprehensive approach to e-waste management.

The bill promotes a policy of the State to protect the environment from the hazards of the improper disposal of obsolete electronic equipment, encourage the responsible manufacturing and sale thereof, and assist consumers in the recycling and disposal of their electronic waste.¹⁷ As of August 30, 2022, this bill is still pending.¹⁸

In my opinion, the delay in passing Senate Bill No. 751 into law is caused by the doubt and possible effects of it on the economy. Particularly establishing the necessary infrastructure for recycling will be costly and enforcing manufacturers' responsibility may exponentially increase its costs which may soon lead to the lack of interest of potential and current investors, from both local and international.

¹⁴ Environment Go; 10 Environmental Impacts of Electronic Waste, Jan. 9, 2023, <https://environmentgo.com/environmental-impacts-of-electronic-waste/>

¹⁵ *Id.*

¹⁶ Senate of the Philippines, 19th Congress; Senate Bill No. 751, <https://legacy.senate.gov.ph/lisdata/3844734904!.pdf>

¹⁷ *Id.*

¹⁸ *Id.*

Other Philippine Statutes Addressing the Issue

R.A. No. 6969 or the Toxic Substances and Hazardous and Nuclear Wastes Control Act of 1990 is a law that aims the regulating, importation, manufacture, handling, and disposal of hazardous substances and waste. It empowers the Department of Environment and Natural Resources (DENR) to oversee toxic chemicals and manage hazardous waste disposal. Furthermore, it also seeks to protect human health and the environment by controlling exposure to toxic chemicals and ensuring proper waste treatment and disposal methods.¹⁹

R.A. No. 9512, known as the *National Environmental Awareness and Education Act of 2008*, aims to increase environmental awareness among Filipinos through education. The law mandates integrating environmental education into school curricula at all levels, including both public and private institutions. It covers topics such as natural resource conservation, waste management, pollution prevention, and climate change. It encourages environmental programs and partnerships among schools, government agencies, and the private sector to foster a culture of environmental protection and sustainable development in the Philippines.²⁰

Both laws are too broad and general in their context. They fail to provide a concrete solution addressing the specific issue of e-waste crisis. The e-waste crisis needs targeted attention as it poses threats to the environment and health. Unlike general solid waste, e-waste contains toxic chemicals and heavy metals, such as mercury, lead, and cadmium, that can leach into soil and water, harming ecosystems and human health. Addressing e-waste separately allows for distinct policies, recycling programs, and handling protocols that would ensure the protection of both public health and the environment more effectively.

State Responsibility of the Philippines

To address the growing threat being imposed by hazardous waste, including electronic waste, the Philippines signed multiple treaties to address the e-waste and one of them is the Basel Convention. Basel Convention, established in 1989 is a global treaty aimed at controlling the transboundary movement and disposal of hazardous wastes, including electronic waste. It requires countries to manage hazardous waste in environmentally sound ways and prohibits the export of such waste to countries lacking the capacity for safe disposal. The convention emphasizes minimizing hazardous waste generation and promotes reducing cross-border waste transfers. It fosters international cooperation and capacity-building to protect human health and the environment from hazardous waste impacts.²¹

¹⁹ Toxic Substances and Hazardous and Nuclear Wastes Control Act of 1990 (Phil.), <https://www.officialgazette.gov.ph/1990/10/26/republic-act-no-6969/>

²⁰ National Environmental Awareness and Education Act of 2008 Rep. Act No. 9512 (Phil.), <https://www.officialgazette.gov.ph/2008/12/12/republic-act-no-9512/>

²¹ Basel Convention 1989 (n.d.), <https://www.basel.int/TheConvention/Overview/tabid/1271/Default.aspx>

The Philippines also has the responsibility to conserve the environment for the present and future generations. Although not yet codified, the principle of intergenerational responsibility to preserve the environment is increasingly recognized as part of customary international law. It is supported by the Stockholm Declaration²² which was later affirmed by the Rio Declaration²³ and enshrined in the UN Framework Convention on Climate Change.²⁴ It reflects a widespread understanding that present generations must act as stewards of the environment for the benefit of future generations, an obligation that many nations integrate into their domestic laws and policies.

As part of its commitment to international environmental standards, the Philippines holds a responsibility to manage and mitigate the impact of e-waste. This involves ensuring environmentally sound disposal of hazardous materials and actively reducing harmful emissions and waste. The nation also upholds an intergenerational duty to protect natural resources, emphasizing sustainable practices that protect the environment for future generations, aligned with both international expectations and national constitutional mandates.

Conclusion

Man has been constantly discovering, inventing, creating, and advancing a lot of things. In the present time, man's capability to transform his surroundings, if used wisely, can be beneficial to people in terms of the benefits of development and the opportunity to enhance the quality of life. Wrongly or heedlessly applied, the same power can do incalculable harm to human beings and the environment. We see around us growing evidence in many regions of the earth of the actual dangerous levels of pollution in living water, air, earth, and beings; major and undesirable disturbances to the ecological balance of the biosphere; destruction and depletion of irreplaceable resources; and gross deficiencies, harmful to the physical, mental and social health of man, in the man-made environment, and working environment.²⁵

The Philippines faces a significant challenge in tackling the growing e-waste crisis. Existing laws fall short of addressing the specific environmental and health threats posed by electronic waste, highlighting the urgent need for more targeted regulations in this time of globalization.

²² Basel Convention 1989 (n.d.), <https://www.basel.int/TheConvention/Overview/tabid/1271/Default.aspx>

²³ Rio Declaration, Report of the United Nations Conference on Environment and Development June 1992 (n.d.), https://www.un.org/en/development/desa/population/migration/generalassembly/docs/globalcompact/A_CONF.151_26_Vol.I_Declaration.pdf

²⁴ United Nations Framework Convention on Climate Change, 1992, <https://unfccc.int/resource/docs/convkp/conveng.pdf>

²⁵ Stockholm Conference, United Nations Conference on the Human Environment, June 1972, <https://documents.un.org/doc/undoc/gen/nl7/300/05/pdf/nl730005.pdf>

A separate law for e-waste, distinct from general solid waste, is indispensable for the reason that electronic waste poses unique challenges and risks not addressed by traditional waste laws. E-waste contains toxic substances like mercury, lead, and cadmium, which can have severe environmental and health damage when improperly disposed of. E-waste requires specialized handling, recycling, and disposal processes to prevent contamination of soil, water, and air. A dedicated law would establish clear guidelines and regulations, focusing on safe e-waste management, producer responsibility, and recycling infrastructure tailored to address the specific hazards and recycling needs of electronic waste. This targeted approach ensures a safer, more effective response to the growing e-waste problem. The enactment of such would also promote accountability among manufacturers, requiring them to take responsibility for e-waste through mechanisms like recycling and safe disposal. Additionally, it would enable a structured, consistent approach to e-waste management nationwide, aligning with international commitments and protecting future generations from suffering long-term ecological harm.

Good intentions alone are insufficient for protecting the environment and addressing e-waste. These efforts must be clearly defined and enforced through well-defined legislation that is binding to everyone.

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